G H Raisoni International Skill Tech University, Pune

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End Semester Examinations - Winter 2025

School of Engineering and Technology

F.Y. B.Tech. Semester - I (All Branches)

Course Name: Indian Knowledge System | **Course Code: 25UHSL1202**

Max Marks: 60 | **Duration: 150 minutes**

Instructions:

1. Write the solution in detail.
2. Clearly show all steps of your work, including necessary derivations and calculations.
3. Read each question carefully before attempting.
4. Students must label their answers with the corresponding question number, such as Q.No. 1A, Q. No. 2C, etc.
5. Each question is of 12 marks, from every question solve any TWO from A, B & C. Question D is compulsory.
6. Calculator is not allowed.

Q. No.	Sub Q.No.	Question	Marks	CO	BL
1		**Solve any TWO from Question A, B & C. Question D is compulsory.**			
	A	What is a Vinculum Number? Convert the following numbers to Vinculum form: a) 67432897 b) 192871	5	CO1	3
	B	Determine the Adjoint of the matrix $A = \begin{bmatrix} 1 & 0 & 5 \\ 2 & 1 & 6 \\ 3 & 4 & 0 \end{bmatrix}$	5	CO1	2
	C	Determine the cofactor of the matrix $A = \begin{bmatrix} 12 & 0 & 5 \\ 23 & 1 & 6 \\ 3 & 7 & 0 \end{bmatrix}$	5	CO1	3
	D	Find cube of: 78 and 35	2	CO1	3
2		**Solve any TWO from Question A, B & C. Question D is compulsory.**			
	A	State the divisibility rule for 4. Check whether the number 65,128 is divisible by 4.	5	CO3	3
	B	Find highest common factor between the polynomials $2x^3 + x^2 - 9$ and $x^4 + 2x^2 + 9$	5	CO3	3
	C	The product of two numbers is 2160 and their GCD is 12. Find their LCM. If one number is 60, find the other number.	5	CO3	3

	D	Define Euclidean algorithm.	2	CO2	1
3		**Solve any TWO from Question A, B & C. Question D is compulsory.**			
	A	Factorise $x^3 + 8x^2 + 19x + 12$.	5	CO3	2
	B	Explain the relationship between roots and coefficients of a polynomial	5	CO3	1
	C	If α, β, γ are the roots of cubic equation $x^3 - 6x^2 + 11x - 6 = 0$, find the value of $\alpha^2 + \beta^2 + \gamma^2$	5	CO3	3
	D	Find the determinant of matrix $\begin{bmatrix} 2 & -3 & 5 \\ 6 & 0 & 4 \\ 1 & 5 & -7 \end{bmatrix}$	2	CO1	1
4		**Solve any TWO from Question A, B & C. Question D is compulsory.**			
	A	Calculate Square root of the following numbers a) 576 b) 9409	5	CO3	3
	B	Using the Vedic method find the Square of 43 and 67	5	CO3	3
	C	State Aryabhata's contribution to the value of π.	5	CO4	3
	D	If one side is given 8 unit, find the other two sides of triangle by using Pythagoras theorem	2	CO3	1
5		**Solve any TWO from Question A, B & C. Question D is compulsory.**			
	A	What is the contribution of Brahmagupta in field of geometry?	5	CO4	1
	B	Bhaskaracharya's *Lilavati* contains a famous problem: "One-fifth of a hive of bees flew to the Kadamba flower..." State the complete problem.	5	CO4	2
	C	Write contribution of Bhaskar Acharya in Mathematics.	5	CO4	3
	D	Name significant contribution of Srinivasa Ramanujan to number theory.	2	CO4	2

- End of the Question Paper -